CBSE CLASS X – MATHEMATICS

MID-TERM EXAMINATION 2026–27

Time Allowed: 3 Hours Maximum Marks: 80

GENERAL INSTRUCTIONS

1. This question paper contains **39 questions**. All questions are compulsory.
2. The question paper is divided into **five sections – A, B, C, D and E**.
3. **Section A** consists of 20 questions of 1 mark each.
4. **Section B** consists of 6 questions of 2 marks each.
5. **Section C** consists of 7 questions of 3 marks each.
6. **Section D** consists of 3 questions of 4 marks each.
7. **Section E** consists of 3 case-study based questions of 5 marks each.
8. Use of calculators is not permitted.
9. Draw neat figures wherever required. Take $\pi = 22/7$ wherever necessary unless stated otherwise.
10. Internal choice is provided in selected questions. Attempt only one option wherever such a choice is given.
11. This paper is based only on the prescribed Mid-Term portion: Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations, Arithmetic Progressions, Triangles, Coordinate Geometry and Areas Related to Circles.

SECTION A – OBJECTIVE TYPE QUESTIONS (20 × 1 = 20)

Q1. Choose the correct option: The HCF of 96 and 404 is:

a) 2 b) 4 c) 8 d) 12

Q2. The decimal expansion of 13/125 is:

a) Non-terminating recurring b) Terminating c) Non-terminating non-recurring d) Irrational

Q3. If α and β are zeroes of $x^2 – 7x + 10$, then $\alpha + \beta$ equals:

a) –7 b) 7 c) 10 d) –10

Q4. A quadratic polynomial whose zeroes are 3 and –2 is:

a) $x^2 + x – 6$ b) $x^2 – x – 6$ c) $x^2 + 5x – 6$ d) $x^2 – 5x + 6$

Q5. The pair of equations $2x + 3y = 6$ and $4x + 6y = 12$ has:

a) Unique solution b) No solution c) Infinitely many solutions d) Exactly two solutions

Q6. The solution of $x + y = 5$ and $x – y = 1$ is:

a) (2, 3) b) (3, 2) c) (1, 4) d) (4, 1)

Q7. The roots of $x^2 – 9 = 0$ are:

a) 3, 3 b) –3, –3 c) 3, –3 d) 9, 1

Q8. For the quadratic equation $x^2 + 4x + 5 = 0$, the nature of roots is:

a) Real and equal b) Real and distinct c) No real roots d) One root is zero

Q9. The 10th term of the AP 3, 7, 11, ... is:

a) 35 b) 39 c) 43 d) 47

Q10. Which term of the AP 5, 9, 13, ... is 49?

a) 10th b) 11th c) 12th d) 13th

Q11. If $\Delta ABC \sim \Delta DEF$ and $AB/DE = 2/3$, then $ar(\Delta ABC) : ar(\Delta DEF)$ is:

a) 2:3 b) 4:9 c) 3:2 d) 9:4

Q12. In ΔABC, if DE ■ BC, then:

a) AD/DB = AE/EC b) AD/AB = EC/AC c) AB/BC = AC/DE d) AD = DB

Q13. The distance between (0, 0) and (6, 8) is:

a) 10 b) 12 c) 14 d) 8

Q14. The midpoint of (2, 4) and (6, 8) is:

a) (4, 6) b) (8, 12) c) (2, 2) d) (3, 4)

Q15. The area of a circle of radius 7 cm is:

a) 44 cm² b) 154 cm² c) 308 cm² d) 22 cm²

Q16. The length of an arc subtending 90° at the centre of a circle of radius 14 cm is:

a) 7 cm b) 14 cm c) 22 cm d) 44 cm

Q17. If the HCF of two numbers is 12 and their LCM is 180, then the product of the numbers is:

a) 192 b) 2160 c) 168 d) 360

Q18. Assertion (A): Every rational number has either a terminating or a non-terminating recurring decimal expansion.
Reason (R): The denominator of a rational number in lowest form contains only prime factors 2 and/or 5 for a terminating decimal expansion.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

Q19. Assertion (A): The graph of a pair of linear equations having a unique solution intersects at exactly one point.
Reason (R): For a unique solution, a■/a■ ≠ b■/b■.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

Q20. If the nth term of an AP is 4n – 1, then its common difference is:

a) 3 b) 4 c) –1 d) 5

SECTION B – VERY SHORT ANSWER QUESTIONS (6 × 2 = 12)

Q21. Use Euclid's division algorithm to find the HCF of 135 and 225.

Q22. Find a quadratic polynomial whose zeroes are 4 and –3.

Q23. Solve: 2x + y = 7 and x – y = 2.

Q24. Find the roots of $x^2 - 5x + 6 = 0$ by factorisation.

Q25. Find the 15th term of the AP: 7, 11, 15, ...

Q26. Find the distance between the points A(–2, 3) and B(4, 3).

SECTION C – SHORT ANSWER QUESTIONS (7 × 3 = 21)

Q27. Prove that $\sqrt{5}$ is irrational.

Q28. If α and β are the zeroes of the polynomial $2x^2 - 7x + 3$, find:
i) $\alpha + \beta$ ii) $\alpha\beta$ iii) $\alpha^2 + \beta^2$

Q29. Solve the pair of linear equations:
3x + 2y = 16
2x – 3y = –1

Q30. Solve $x^2 - 7x + 10 = 0$ using the quadratic formula.

Q31. The 7th term of an AP is 19 and the 13th term is 37. Find the AP.

Q32. In ∆ABC, DE ■ BC, where D lies on AB and E lies on AC. If AD = 3 cm, DB = 2 cm and AE = 4.5 cm, find EC. Give reasons.

Q33. Find the area of the sector of a circle of radius 21 cm whose central angle is 120°. Also find the length of its arc.

SECTION D – LONG ANSWER QUESTIONS (3 × 4 = 12)

Q34. A fraction becomes 1/3 when 1 is subtracted from its numerator and denominator. It becomes 1/2 when 2 is added to both numerator and denominator. Find the fraction.
OR

The sum of two numbers is 27 and their difference is 5. Find the numbers using a pair of linear equations.

Q35. The area of a right triangle is 60 cm². If its base is 7 cm more than its height, find the base and height using a quadratic equation.

OR

The product of two consecutive positive integers is 306. Find the integers.

Q36. In ΔABC, DE ■ BC. If AD = 4 cm, DB = 6 cm and area of ΔADE = 24 cm², find:

i) ar(ΔABC)

ii) ar(trapezium DBCE)

iii) ar(ΔADE) : ar(ΔABC)

SECTION E – CASE STUDY BASED QUESTIONS (3 × 5 = 15)

Q37. CASE STUDY: ARITHMETIC PROGRESSION

A school auditorium has seats arranged in rows. The first row has 20 seats, the second row has 24 seats, the third row has 28 seats and so on. Thus, the number of seats in successive rows forms an AP.

i) Write the first term and common difference of the AP. (1)

ii) Find the number of seats in the 10th row. (1)

iii) Which row has 80 seats? (1)

iv) Find the total number of seats in the first 10 rows. (2)

Q38. CASE STUDY: COORDINATE GEOMETRY

Three friends are standing at points A(–3, 2), B(5, 2) and C(1, 8) on a coordinate plane, forming a triangular region.

i) Find the distance AB. (1)

ii) Find the midpoint of AB. (1)

iii) Find the distance between A and C. (1)

iv) Show that the triangle is isosceles or not by comparing AC and BC. (2)

Q39. CASE STUDY: AREAS RELATED TO CIRCLES

A circular park has radius 14 m. A sector of 90° is marked for a flower garden and the remaining part is used as a lawn. Take π = 22/7.

i) Find the circumference of the park. (1)

ii) Find the area of the entire park. (1)

iii) Find the area of the 90° sector. (1)

iv) Find the area of the remaining lawn. (2)

— END OF QUESTION PAPER —

QUESTION PAPER BLUEPRINT

Type of Questions	No. of Questions	Total Marks
1 Mark Questions	20	20 × 1 = 20
2 Mark Questions	6	6 × 2 = 12
3 Mark Questions	7	7 × 3 = 21
4 Mark Questions	3	3 × 4 = 12
5 Mark Questions	3	3 × 5 = 15
Total	**39**	**80**